Ohm’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect voltage, current and resistance for an ohmic component.

SUCCESS CRITERIA - I can...

1. Identify the voltage across a resistor.
2. Use V=IR with appropriate units.
3. Predict changes while stating what stays fixed.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ohmic” and “resistance” to explain your answer. Sentence starters are optional.

ohmic: having constant resistance under the stated physical conditions

resistance: voltage divided by current for a stated operating condition

voltage: electric potential difference; energy transferred per charge

01 Warm-up: bring back an earlier idea

Move from r to **3** r away from a point charge. **Compare field magnitude** and say what you treated as true about the source charge.

Word help: charge = a property responsible for electric interactions

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: V = IR; I = V/R.
Ohmic resistor with temperature and other physical properties kept the same. Resistance can change with temperature.

Worked reasoning: Across a **4 ohm** resistor, a **12 V** supply gives I=V/R=**3 A**. Doubling resistance at the same voltage halves current.

Your turn: Draw a one-resistor circuit, label supply voltage and current, and **mark both resistor terminals**.

Sentence starter: In my work, I used … to show … . I labeled it … .



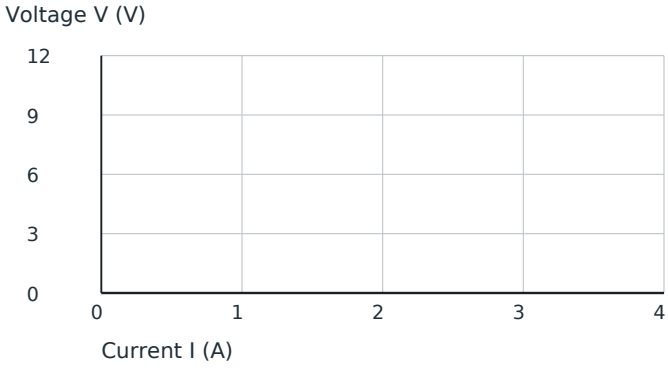
Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch a graph of V** against I for a fixed ohmic resistor.
2. What does its slope represent?

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

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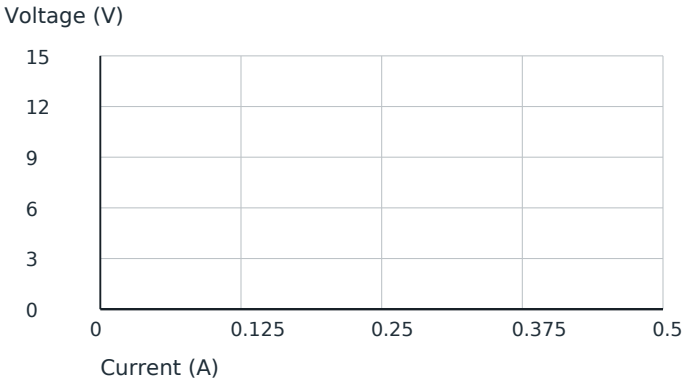
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A resistor has **12 V** across its terminals and carries **0.40 A**. **Find resistance**. If the voltage becomes **9 V** and resistance stays constant, find the new current. **Label both operating points** on the graph.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

A student doubles resistance while keeping voltage fixed and predicts twice the current. Correct the prediction.

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An EM wave in vacuum has frequency **6×10^{9} Hz**. Use c=**3.00×10^{8} m/s**. **Find wavelength** and **name the spectral region**.
Label a wave sketch to **support the calculation or comparison**.

Word help: vacuum = space with negligible matter

Sentence starter: I found ... [unit]. This value means ... in this situation.



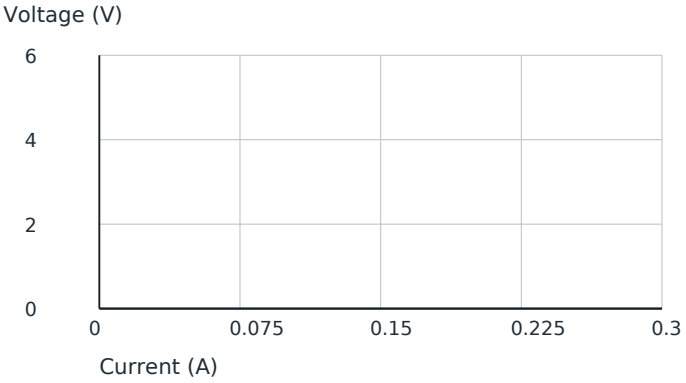
Time or position: choose and label

Marzano check - circle: ① ② ③

07 Exit check

Measured values are (I, V)=(**0.10**, **2.0**),(**0.20**, **4.0**),(**0.30**, **6.0**). **Find resistance** and **explain what supports treating the** resistance as constant.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

An ohmic component has **4 V** across it at **0.5 A**.
1. **Find resistance** and current at **10 V**.
2. **Explain why this extrapolation requires** the component to remain ohmic.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③